

Task 2: Transfer Learning using ResNet50

Performance Comparison Report — Baseline CNN vs. Pretrained ResNet50

1. Objective

To implement transfer learning using a pre-trained deep learning model (ResNet50) and analyze performance improvements over a custom CNN architecture, using the Flowers Recognition dataset.

2. Dataset

- Dataset: Flowers Recognition Dataset (Kaggle — alxmamaev/flowers-recognition)
- Classes (5): daisy, dandelion, rose, sunflower, tulip
- Total images: 4317 | Training: 3454 | Validation: 863 (80/20 split)
- Image size: 180 x 180 | Batch size: 32

3. Approach

- Load and preprocess the dataset (resize, rescale, augment).
- Build and train a baseline custom CNN from scratch (3 conv blocks + dense classifier).
- Load a pretrained ResNet50 (ImageNet weights), freeze the base layers, and add a new classification head.
- Train both models for 10 epochs each and record training time.
- Compare validation accuracy, validation loss, and training time.
- Visualize and summarize results and observations.

4. Source Code

4.1 Imports & Dataset Download

```
import os
import time
import matplotlib.pyplot as plt
import numpy as np
import tensorflow as tf
from tensorflow.keras import layers, models, applications, optimizers

import kagglehub

path = kagglehub.dataset_download("alxmamaev/flowers-recognition")
print("Path to dataset files:", path)

dataset_dir = os.path.join(path, "flowers") if os.path.exists(os.path.join(path,
"flowers")) else path
```

Output

```
Downloading to /root/.cache/kagglehub/datasets/alxmamaev/flowers-recognition/2.archive...
100%|#####| 225M/225M [00:12<00:00, 19.2MB/s]
Extracting files...
Path to dataset files: /root/.cache/kagglehub/datasets/alxmamaev/flowers-recognition/versions/2
```

4.2 Data Loading & Preprocessing

```
# 2. Data loading and Pre Processing
IMG_SIZE = (180, 180)
BATCH_SIZE = 32

# Load train and validation splits using image_dataset_from_directory
train_ds = tf.keras.utils.image_dataset_from_directory(
    dataset_dir,
    validation_split=0.2,
    subset="training",
    seed=123,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

val_ds = tf.keras.utils.image_dataset_from_directory(
    dataset_dir,
    validation_split=0.2,
    subset="validation",
    seed=123,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

class_names = train_ds.class_names
num_classes = len(class_names)
print(f"Classes found ({num_classes}): {class_names}")

# Performance optimization: Prefetching & Data Rescaling Layer
rescale_layer = layers.Rescaling(1./255)

train_ds = train_ds.map(lambda x, y: (rescale_layer(x),
y)).prefetch(buffer_size=tf.data.AUTOTUNE)
val_ds = val_ds.map(lambda x, y: (rescale_layer(x),
```

```
y)).prefetch(buffer_size=tf.data.AUTOTUNE)

# Data Augmentation Pipeline
data_augmentation = models.Sequential([
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.15),
    layers.RandomZoom(0.15),
], name="data_augmentation")
```

Output

```
Found 4317 files belonging to 5 classes.
Using 3454 files for training.
Found 4317 files belonging to 5 classes.
Using 863 files for validation.
Classes found (5): ['daisy', 'dandelion', 'rose', 'sunflower', 'tulip']
```

4.3 Baseline CNN Model

```
def build_baseline_cnn():
    model = models.Sequential([
        layers.Input(shape=(180, 180, 3)),
        data_augmentation,
        layers.Conv2D(32, (3, 3), activation='relu'),
        layers.MaxPooling2D((2, 2)),
        layers.Conv2D(64, (3, 3), activation='relu'),
        layers.MaxPooling2D((2, 2)),
        layers.Conv2D(128, (3, 3), activation='relu'),
        layers.MaxPooling2D((2, 2)),
        layers.Flatten(),
        layers.Dense(128, activation='relu'),
        layers.Dropout(0.5),
        layers.Dense(num_classes, activation='softmax')
    ], name="Baseline_CNN")

    model.compile(
        optimizer='adam',
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )
    return model
```

4.4 Transfer Learning Model (ResNet50)

```
def build_transfer_learning_resnet():
    # Load base model with ImageNet weights (excluding top classification head)
    base_model = applications.ResNet50(
        weights='imagenet',
        include_top=False,
        input_shape=(180, 180, 3)
    )

    # Freeze base feature extractor
    base_model.trainable = False

    inputs = layers.Input(shape=(180, 180, 3))
    x = data_augmentation(inputs)
    # Scaled back to [0, 255] for ResNet internal preprocessing requirement
    x = applications.resnet.preprocess_input(x * 255.0)
    x = base_model(x, training=False)
    x = layers.GlobalAveragePooling2D()(x)
```

```
x = layers.BatchNormalization()(x)
x = layers.Dense(256, activation='relu')(x)
x = layers.Dropout(0.4)(x)
outputs = layers.Dense(num_classes, activation='softmax')(x)

model = models.Model(inputs, outputs, name="ResNet50_TransferLearning")

model.compile(
    optimizer=optimizers.Adam(learning_rate=1e-3),
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
return model
```

4.5 Model Training

```
EPOCHS = 10

# Train Custom Baseline CNN
print("\n" + "="*50)
print(" Training Baseline Custom CNN ")
print("="*50)
baseline_cnn = build_baseline_cnn()
start_cnn = time.time()
history_cnn = baseline_cnn.fit(train_ds, validation_data=val_ds, epochs=EPOCHS)
time_cnn = time.time() - start_cnn

# Train ResNet50 Transfer Learning Model
print("\n" + "="*50)
print(" Training ResNet50 Transfer Learning Model ")
print("="*50)
resnet_tl = build_transfer_learning_resnet()
start_tl = time.time()
history_tl = resnet_tl.fit(train_ds, validation_data=val_ds, epochs=EPOCHS)
time_tl = time.time() - start_tl
```

Output — Baseline CNN Training Log

```
=====
Training Baseline Custom CNN
=====
Epoch 1/10 108/108 13s 58ms/step - accuracy: 0.4218 - loss: 1.3161 - val_accuracy: 0.5632 -
val_loss: 1.0848
Epoch 2/10 108/108 7s 62ms/step - accuracy: 0.5388 - loss: 1.1379 - val_accuracy: 0.6118 -
val_loss: 0.9819
Epoch 3/10 108/108 6s 54ms/step - accuracy: 0.5860 - loss: 1.0372 - val_accuracy: 0.6385 -
val_loss: 0.9629
Epoch 4/10 108/108 10s 55ms/step - accuracy: 0.6190 - loss: 0.9776 - val_accuracy: 0.6628 -
val_loss: 0.8759
Epoch 5/10 108/108 6s 51ms/step - accuracy: 0.6265 - loss: 0.9481 - val_accuracy: 0.6756 -
val_loss: 0.8459
Epoch 6/10 108/108 10s 51ms/step - accuracy: 0.6584 - loss: 0.9004 - val_accuracy: 0.6767 -
val_loss: 0.8175
Epoch 7/10 108/108 7s 63ms/step - accuracy: 0.6824 - loss: 0.8567 - val_accuracy: 0.6976 -
val_loss: 0.7811
Epoch 8/10 108/108 11s 68ms/step - accuracy: 0.6711 - loss: 0.8452 - val_accuracy: 0.7045 -
val_loss: 0.7993
Epoch 9/10 108/108 5s 50ms/step - accuracy: 0.6943 - loss: 0.8156 - val_accuracy: 0.7057 -
val_loss: 0.7886
Epoch 10/10 108/108 7s 62ms/step - accuracy: 0.7038 - loss: 0.7937 - val_accuracy: 0.6906 -
val_loss: 0.8023
```

Output — ResNet50 Transfer Learning Training Log

```
=====
Training ResNet50 Transfer Learning Model
=====
Downloading resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5 ... 94765736/94765736 6s
0us/step
Epoch 1/10 108/108 25s 151ms/step - accuracy: 0.7655 - loss: 0.8050 - val_accuracy: 0.8563 -
val_loss: 0.4367
Epoch 2/10 108/108 13s 120ms/step - accuracy: 0.8379 - loss: 0.4957 - val_accuracy: 0.8575 -
val_loss: 0.4143
Epoch 3/10 108/108 13s 121ms/step - accuracy: 0.8796 - loss: 0.3512 - val_accuracy: 0.8633 -
val_loss: 0.4270
Epoch 4/10 108/108 13s 123ms/step - accuracy: 0.8885 - loss: 0.3028 - val_accuracy: 0.8679 -
val_loss: 0.4475
Epoch 5/10 108/108 13s 123ms/step - accuracy: 0.9007 - loss: 0.2707 - val_accuracy: 0.8737 -
val_loss: 0.4658
Epoch 6/10 108/108 13s 124ms/step - accuracy: 0.9155 - loss: 0.2264 - val_accuracy: 0.8772 -
```

```

val_loss: 0.4701
Epoch 7/10 108/108 13s 122ms/step - accuracy: 0.9189 - loss: 0.2220 - val_accuracy: 0.8783 -
val_loss: 0.4435
Epoch 8/10 108/108 13s 122ms/step - accuracy: 0.9236 - loss: 0.2150 - val_accuracy: 0.8795 -
val_loss: 0.4595
Epoch 9/10 108/108 13s 122ms/step - accuracy: 0.9285 - loss: 0.2071 - val_accuracy: 0.8830 -
val_loss: 0.4446
Epoch 10/10 108/108 13s 122ms/step - accuracy: 0.9369 - loss: 0.1791 - val_accuracy: 0.8922 -
val_loss: 0.4326

```

4.6 Accuracy & Loss Visualization

```

# 6. Accuracy & Loss Visualization Plots

plt.figure(figsize=(15, 5))

# Validation Accuracy Plot
plt.subplot(1, 2, 1)
plt.plot(history_cnn.history['val_accuracy'], label='Baseline CNN Val Acc',
color='crimson', linestyle='--', linewidth=2)
plt.plot(history_tl.history['val_accuracy'], label='ResNet50 Val Acc', color='teal',
linewidth=2)
plt.title('Validation Accuracy Comparison')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend(loc='lower right')
plt.grid(True)

# Validation Loss Plot
plt.subplot(1, 2, 2)
plt.plot(history_cnn.history['val_loss'], label='Baseline CNN Val Loss',
color='crimson', linestyle='--', linewidth=2)
plt.plot(history_tl.history['val_loss'], label='ResNet50 Val Loss', color='teal',
linewidth=2)
plt.title('Validation Loss Comparison')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend(loc='upper right')
plt.grid(True)

plt.tight_layout()
plt.show()

```

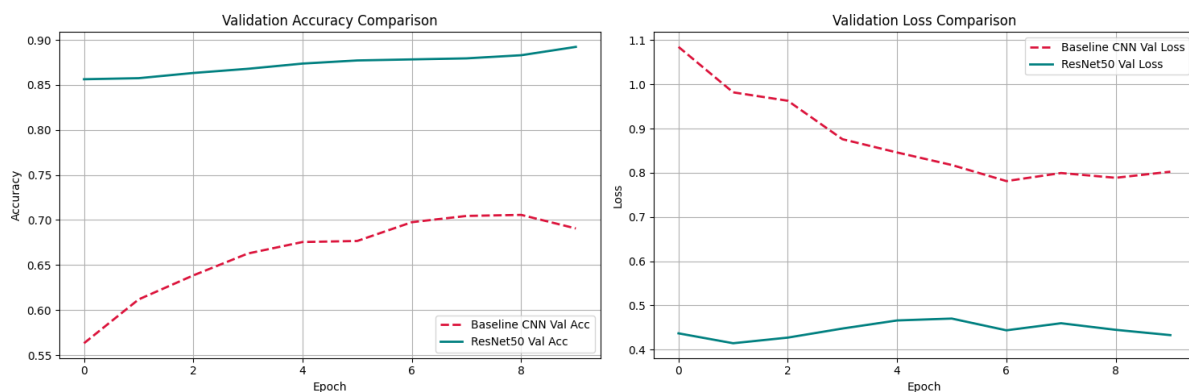


Figure: Validation accuracy and validation loss comparison — Baseline CNN vs. ResNet50

4.7 Performance Summary Table (code)

```
# 7. Print Performance Summary Table

val_acc_cnn = max(history_cnn.history['val_accuracy']) * 100
val_acc_tl = max(history_tl.history['val_accuracy']) * 100
val_loss_cnn = min(history_cnn.history['val_loss'])
val_loss_tl = min(history_tl.history['val_loss'])

print("\n" + "="*60)
print(f"{'Metric':<25} | {'Baseline CNN':<15} | {'ResNet50 TL':<15}")
print("="*60)
print(f"{'Best Val Accuracy (%)':<25} | {val_acc_cnn:<15.2f} | {val_acc_tl:<15.2f}")
print(f"{'Lowest Val Loss':<25} | {val_loss_cnn:<15.4f} | {val_loss_tl:<15.4f}")
print(f"{'Total Time (seconds)':<25} | {time_cnn:<15.2f} | {time_tl:<15.2f}")
print("="*60)
```

Output

```
=====
Metric                | Baseline CNN      | ResNet50 TL
=====
Best Val Accuracy (%) | 70.57             | 89.22
Lowest Val Loss       | 0.7811            | 0.4143
Total Time (seconds)  | 81.09             | 143.83
=====
```

5. Results

The table below summarizes the final performance comparison between the baseline custom CNN and the ResNet50-based transfer learning model, both trained for 10 epochs on the same train/validation split.

Metric	Baseline CNN	ResNet50 (Transfer Learning)
Best Validation Accuracy	70.57%	89.22%
Lowest Validation Loss	0.7811	0.4143
Validation Accuracy (Epoch 1)	56.32%	85.63%
Total Training Time (10 epochs)	81.09 s	143.83 s

6. Observations

1. Accuracy improvement: The ResNet50 transfer learning model achieved a best validation accuracy of 89.22%, an improvement of roughly 18.65 percentage points over the baseline CNN's 70.57%. This confirms that features learned from ImageNet transfer effectively to the flower classification task.

2. Faster convergence: The transfer learning model reached 85.63% validation accuracy in just the first epoch — a level the baseline CNN never reached even after 10 full epochs. Pretrained convolutional filters (edges, textures, shapes) already encode useful representations, drastically reducing the epochs needed to converge.

3. Lower and more stable loss: The final validation loss for the transfer learning model (0.4143) is less than half that of the baseline CNN (0.7811), indicating more confident and better-calibrated predictions.

4. Baseline CNN limitations: Training from scratch on a relatively small dataset (~3454 training images) causes the baseline CNN to plateau around 70% accuracy, with a noisier validation curve (e.g., val_loss ticks up slightly at epoch 8 before recovering) due to limited learned generalizable features.

5. Training time trade-off: The transfer learning model took about 1.77x longer per run (143.83s vs 81.09s for 10 epochs), mainly due to the larger ResNet50 backbone's forward pass on every batch. However, given the large accuracy gain and much faster convergence, far fewer epochs would be needed in practice, making transfer learning more time-efficient overall for reaching a target accuracy.

6. Overfitting control: The transfer learning model's training accuracy (93.69% by epoch 10) stayed reasonably close to its validation accuracy (89.22%), showing that freezing the base layers, dropout (0.4), and data augmentation together controlled overfitting effectively.

7. Conclusion

Transfer learning with a pretrained ResNet50 clearly outperforms a custom CNN trained from scratch on the Flowers Recognition dataset — delivering higher accuracy, lower loss, and dramatically faster convergence, at the cost of a moderate increase in per-epoch training time. For small-to-medium sized image classification datasets, transfer learning is the more effective and practical approach compared to training a CNN from scratch.

Github:

https://github.com/AdityaBahira/Lnt_Deep_Learning/tree/main/Task%20%3A%20Transfer%20Learning%20using%20VGG16